The Role of Barotropic versus Baroclinic Feedbacks on the Eddy Response to Annular Mode Zonal Wind Anomalies

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ABSTRACT: The annular mode, the leading pattern of low frequency variability in the extratropics, owes its temporal persistence to a positive feedback between eddy momentum fluxes and the background zonal wind anomalies associated with the annular mode itself. The mechanisms by which the zonal wind anomalies impact the eddy momentum fluxes fall into two families: 1) baroclinic mechanisms: changes in the amount and location of wave activity generated via baroclinic instability causes the changes in eddy momentum fluxes and 2) barotropic mechanisms: the zonal wind anomalies impact the eddy momentum fluxes directly via critical levels, turning latitudes and the refraction of meridionally propagating waves. This paper takes a critical look at various methodologies that conclude that baroclinic feedbacks are dominant by developing multiple independent estimates of the relative role of baroclinic versus barotropic processes. All methods conclude that barotropic mechanisms are most important, however, baroclinic mechanisms are not negligible. Additional experiments with the baroclinic feedback turned off (via manipulations to the vertical friction profile) also suggest that barotropic feedbacks are dominant. The methods for estimating the feedbacks are: 1) Rossby Wave Chromatography, 2) forced manipulations of the vertical structure of EOF1 using Linear Response Functions and 3) quantitatively inferring the meridional wave propagation from the mean wave activity budget and then using this to analyze the wave activity response to anomalies. The last method is also applied to both Northern and Southern Hemisphere reanalysis and similar conclusions regarding the feedbacks are reached.

1. Introduction

The annular mode, which is leading pattern of low frequency variability in the extratropics (Thompson and Wallace (2000)), has a wide range of important societal and climatic impacts including surface temperature and precipitation (Gillett et al. (2006); Hendon et al. (2007); Hendon et al. (2014)), blocking and cold air outbreaks (Kaas and Branstator (1993); Thompson and Wallace (2001)), orographic precipitation (Shi and Durran (2014)), the earth radiation budget via cloudiness (Grise et al. (2013); Li et al. (2014)), the ocean circulation and sea ice (Hall and Visbeck (2002); Delworth and Zeng (2008); Spence et al. (2014)) and ocean biology (Lovenduski and Gruber (2005)). The annular mode is closely related to the leading EOF of zonal-mean zonal wind (U) and previous studies (Lorenz and Hartmann (2001); Lorenz and Hartmann (2003)) have used this U EOF to understand annular dynamics. In comprehensive climate models, annular variability is often not well represented. In particular, the persistence of U anomalies is typically too large in models (Gerber et al. (2008); Kidston and Gerber (2010)). It is generally agreed that a positive feedback between eddy momentum fluxes and U anomalies is a major contributor to annular mode persistence (Robinson (1996); Lorenz and Hartmann (2001); Lorenz and Hartmann (2003); Simpson et al. (2013)). Therefore, an understanding of the mechanisms of the positive feedback is very important for understanding model biases. The effect of eddy momentum fluxes on U is well understood via the zonal-mean momentum budget. It is much more difficult to understand the effect of U on the eddies, which closes the positive feedback loop.

The mechanisms by which U affects the eddy momentum fluxes fall into two families: "baroclinic" and "barotropic". For the baroclinic mechanism, it is hypothesized that U anomalies cause collocated anomalies of baroclinic instability of the same sign. The positive baroclinic instability anomalies are equivalent to increases in the upward flux of wave activity into the upper troposphere (Edmon et al. (1980)). Baroclinic mechanisms argue that increases in the baroclinic wave source leads to increases in the wave activity diverging meridionally at upper levels, which is equivalent to reinforcing eddy momentum flux convergence at upper levels (Edmon et al. (1980)). Papers that advocate a baroclinic mechanism include Robinson (1996), Robinson (2000), Lorenz and Hartmann (2001), Chen and Plumb (2009), Rivière (2009), Zhang et al. (2012) and Zurita-Gotor et al. (2014).

For barotropic mechanisms, it is hypothesized that the U anomalies change the meridional wave propagation (i.e. eddy momentum flux) directly, independent of changes in the baroclinic wave source. This could come about through linear or nonlinear mechanisms. For example, U directly impacts the location of upper-level critical levels where Rossby waves decay, thus directly impacting the eddy momentum flux divergence (Barnes et al. (2010), Barnes and Hartmann (2011) and Lorenz (2014b)). In addition, U directly impacts the location of Rossby wave turning latitudes, which impacts the direction of meridional wave propagation (Barnes and Hartmann (2011); Lorenz (2014b)). Additional papers that advocate barotropic mechanisms include Hartmann (1995), Hartmann and Zuercher (1998), Jin et al. (2006a), Jin et al. (2006b) and Chen and Zurita-Gotor (2008).

In most cases, baroclinic mechanisms have been advocated using the anomalous upper-level wave activity budget (Lorenz and Hartmann (2001), Zurita-Gotor et al. (2014), Nie et al. (2014)). If the convergence of the vertical EP flux in the upper troposphere is greater than or equal to the divergence of the horizontal EP flux, then this is considered sufficient evidence that baroclinic feedbacks are operating. The problem with this approach is that the ratio of the vertical to horizontal EP flux of the *mean* state is not taken into account (Robert et al. (2017)). In the mean state, the vertical EP flux is typically at least several times greater than the horizontal EP flux (e.g. Edmon et al. (1980)), which implies that most baroclinic wave activity does not propagate meridionally, instead the wave activity dissipates locally. If anomalies behave like the mean state, then the anomalous baroclinic wave source must be significantly larger than the divergence of the horizontal EP flux (i.e. eddy momentum flux convergence) for the baroclinic feedback to fully account for the feedback. Robert et al. (2017) first advocated these ideas and coined the term "radiation efficiency", which is the ratio of horizontal EP flux divergence to vertical EP flux convergence in the upper troposphere in the mean state, to better quantify the baroclinic feedback.

Alternatively, Hassanzadeh and Kuang (2019) advocated baroclinic mechanisms by manipulating the vertical structure of EOF1 and observing the eddy response. When EOF1 is made barotropic, the eddy momentum flux no longer reinforces the U anomalies, which is consistent with a baroclinic feedback. However, the barotropic U anomalies do not give zero baroclinic source anomalies (see heat flux response in Hassanzadeh and Kuang (2019)), instead the baroclinic source anomalies are strong but in the opposite direction of the standard EOF1 case. In this paper, we repeat

the Hassanzadeh and Kuang (2019) experiments under the assumption that both barotropic and baroclinic mechanisms are operating. With two mechanisms and two experiments (standard EOF and barotropic EOF), one can linearly disentangle the contribution of each mechanism. Under this more general framework, we find that the barotropic mechanism is most important but that the baroclinic feedback is non-negligible.

Unlike all studies advocating baroclinic mechanisms, some studies using barotropic methods go beyond simply diagnosing the source of the feedbacks. They also explicitly predict either the structure of of the leading mode (Jin et al. (2006a); Jin et al. (2006b)) or the latitudinal structure of positive and/or propagating eddy feedbacks (Lorenz (2014a); Lorenz (2014b); Lorenz (2015)). The former studies predict annular mode structure from the mean state and the space-time structure of the transient eddy field. The latter studies predict the eddy momentum fluxes from the mean state and the phase-speed/latitude structure of the baroclinic wave source (convergence of the vertical EP flux in the upper troposphere). Lorenz (2015) called their method Rossby Wave Chromatography (RWC) (Held and Phillips (1987)). All these studies assume linear dynamics for the eddy field.

The goal of this paper is not to provide a "best estimate" of the relative role of baroclinic versus barotropic feedbacks in a specific model and/or reanalysis. Instead, the goal is to critically assess certain methodologies that have been used in the past to conclude that baroclinic feedbacks are dominant. In particular, we believe that studies that diagnose feedbacks 1) from the anomalous upper-level wave activity budget (Lorenz and Hartmann (2001), Zurita-Gotor et al. (2014), Nie et al. (2014)) or 2) via manipulations of the vertical structure of EOF1 (Hassanzadeh and Kuang (2019)) have overestimated the baroclinic feedback in the past. To support our hypotheses, we develop several independent methodologies to estimate the relative contribution of barotropic and baroclinic feedbacks. In order to maximize the ability to probe the dynamics, this paper focuses on General Circulation Models (GCMs) with simplified physics. First, we provide an experiment which demonstrates that a naive analysis of the upper level wave activity budget is misleading by turning off the baroclinic feedback via modifications to the Rayleigh friction profile. Next, we estimate the barotropic and baroclinic feedbacks using RWC (Lorenz (2015)) and an updated baroclinic RWC that is presented below. Then we provide independent estimates of the feedbacks using an alternate interpretation of the Linear Response Function (LRF) method of Hassanzadeh and Kuang (2016) and Hassanzadeh and Kuang (2019). Next we develop a simple model that

infers the amount of wave propagation from the time-mean wave activity budget. Applying this Inferred Wave Propagation (IWP) model to anomalies provides another estimate of barotropic and baroclinic feedbacks. Finally the IWP model is applied to reanalysis to estimate feedbacks. All methods estimate that barotropic feedbacks are more important than baroclinic feedbacks for the GCM (best guess: 70% barotropic and 30% baroclinic). For observations, the estimates are seasonally dependent with barotropic feedbacks ranging from 60% to 100% of the total feedback.

2. Methods

In this paper, we will be applying three types of analysis methodologies to a hierarchy of two GCMs. In this section, we first describe the two GCMs. Next, we describe two of the three methods. The final method is explained in Section 3d. The final method will also be applied to reanalysis.

a. Models

The are two GCMs used in this study, and both are idealized dry dynamical cores. The difference between the GCMs is the number of vertical levels. One is a standard multi-level primitive equation model and the second is a two-level primitive equation model. The multi-level is only used in Section 3a.

1) Multi-level GCM

The multi-level GCM is a standard primitive equation spectral model on a sphere integrating the vorticity, divergence, temperature and the log surface pressure. The sigma coordinate vertical differencing scheme of Simmons and Burridge (1981) is used. An 8th order hyperdiffusion with a time scale of 0.1 days for the smallest scale waves is applied to the model variables. The only non-typical aspect of the model is the time differencing, which is the AB3-AI2 method of Durran and Blossey (2012) rather than Leapfrog-Trapezoidal differencing. The explicit component of this scheme is the third-order Adams-Bashforth method, which has distinct advantages over the leapfrog scheme (Durran (1991)). The model was independently coded and verified with the Held and Suarez (1994) and Polvani et al. (2004) numerical tests. The resolution of all simulations is T42 with 20 equally spaced vertical levels. The control simulation is forced with the diabatic

heating and frictional damping of Held and Suarez (1994). For each experiment, the model is run for 6500 days and the first 500 days are discarded to allow for model spin-up.

2) Two-level GCM

For efficient implementation of the many long-term-integrations required for Linear Response Function (LRF) analysis (Hassanzadeh and Kuang (2016)), we also use a two-level primitive equation model (GCM) based on Hendon and Hartmann (1985). The two-level model is also used for the Rossby Wave Chromatography analysis (Lorenz (2015)). The prognostic variables of vorticity, divergence and potential temperature are defined on two presssure levels (250 and 750hPa). In the equations, the vertically integrated divergence is constrained to be zero for consistency with the assumption of constant surface pressure at the lower boundary. For thermal forcing we use a Newtonian relaxation of the temperature field to the Held and Suarez (1994) equilibrium temperature at 250 and 750hPa, with a thermal damping time scale of 40 days. For mechanical damping we use Rayleigh friction at the lower-level with an e-folding time scale of 3 days. In the past, this model has been run at coarse resolution (R15) with semi-implicit time differencing (Hendon and Hartmann (1985); Robinson (1991)). At the T42 resolution used here, however, the speed of the winds is approximately equal to the phase speed of the fastest gravity mode (which is slower than usual due to the constant surface pressure) and therefore our version is fully explicit with third-order Adams-Bashforth time differencing (Durran (1991)). This is not a concern because the purpose of the semi-implicit method used in multi-level, spectral GCMs is also to reduce gravity wave phase speeds. For each experiment, the model is run for 251000 days and the first 1000 days are discarded to allow for model spin-up. The long integration length is required for the LRF analysis of Hassanzadeh and Kuang (2016).

For the multi-level GCM, EP fluxes (Edmon et al. (1980)) are calculated using straightforward second-order, centered finite differences. For the two-level primitive equation model, however, the calculation of EP fluxes requires more care. Because the Edmon et al. (1980) EP fluxes assume quasi-geostrophy, we formulate the vertical EP flux convergence so that it is most consistent with the analogous two-level quasi-geostrophic model. Therefore, the vertical EP flux convergence at

the upper-level (i.e. the baroclinic wave source) of the two-level primitive equation model is:

$$\text{source} = \frac{\cos\phi f \overline{\theta' v'}}{\frac{1}{2}(\overline{\theta}_1 - \overline{\theta}_2)}, \tag{1}$$

where θ is the potential temperature, ϕ is the latitude, f is the Coriolis parameter, θ' is the eddy component of $\frac{\theta_1+\theta_2}{2}$, v' is the eddy component of $\frac{v_1+v_2}{2}$, the overbar is the zonal-mean, level 1 is the upper level, level 2 is the lower level and the $\frac{1}{2}$ factor in the denominator is needed for consistency between two-level quasi-geostrophic and primitive equation vertical differencing.

b. Phase Speed/Latitude Spectra

Space-time cross-spectral analyses of eddy fluxes are calculated using the method of Randel and Held (1991). The temporal spectral analysis is performed over 64 day chunks (= 192 times given the 8 hour sampling time) that overlap by 32 days. The resulting frequency spectrum, which is linearly spaced in frequency, is further smoothed with a running mean over five adjacent frequency bands. The phase speed spectra are given in terms of angular phase speed multiplied by $a\cos(45°)$, where a is the radius of the earth so that the angular phase speed is approximately equal to the phase speed in the mid-latitude. Typically, the stationary phase speed/frequency is not considered because the effects of strong stationary wave sources dominate the observed phase speed spectra. In our models with axisymmetric boundary conditions, however, the stationary frequency (phase speed) is not enhanced relative to adjacent frequency bands and we therefore consider the full phase speed spectrum.

c. Rossby Wave Chromatography (RWC)

1) Barotropic RWC

Barotropic RWC (Lorenz (2015)) calculates the eddy momentum fluxes from the zonal-mean zonal wind (U) and the space-time structure of the baroclinic wave source. Because U and the baroclinic wave source are prescribed independently, the contribution of each can be isolated. RWC is based on the barotropic vorticity equation on a sphere linearized about a background zonal-mean

zonal wind, $\bar{u}$, and absolute vorticity gradient, β^* $(= a^{-1}\partial_\phi(f+\bar{\zeta}))$,

$$\frac{\partial \zeta'}{\partial t} + \frac{\bar{u}}{a\cos\phi}\frac{\partial \zeta'}{\partial \lambda} + \beta^* v' - \nu\nabla^2\zeta' = F'_\zeta, \tag{2}$$

where ζ', v' and F'_ζ are the eddy relative vorticity, meridional velocity and vorticity forcing, λ is the longitude, a is the radius of the earth, ν is the diffusion coefficient. The forcing, F'_ζ, represents the input of wave activity via baroclinic instability and the diffusive damping is a parameterization of wave dissipation via nonlinear wave breaking. Forced barotropic and shallow water models have been used before to understand zonal-mean zonal wind variability and change (Vallis et al. (2004); Chen et al. (2007); Barnes et al. (2010); Barnes and Hartmann (2011); Kidston and Vallis (2012)). Unlike these studies, which specify the forcing directly, RWC prescribes the wave activity source and then determines the forcing, vorticity and all other fields from the wave source under the assumption that the forcing and vorticity are related by the linearized barotropic vorticity equation (Lorenz (2015)). Note, RWC directly calculates the full fields, not the perturbation fields. The advantages of prescribing the wave source are discussed in Lorenz (2014a); Lorenz (2014b). As in Lorenz (2015), we use the convergence of vertical EP flux (Edmon et al. (1980)) in the upper troposphere for the "baroclinic" wave source and, because the background flow is steady and zonally symmetric, RWC is applied separately to each zonal wavenumber and phase speed (Randel and Held (1991)). The model first predicts the eddy vorticity field, which then allows the eddy vorticity and momentum fluxes to be calculated.

With RWC, the baroclinic wave activity source and the background flow are decoupled so that each can be manipulated separately to better understand the dynamics. Here, the baroclinic feedback is defined as the response of $\overline{u'v'}$ to changes in the amplitude of the total baroclinic wave activity source integrated over phase speed. The barotropic feedback is defined as the response to changes in the background flow *and* changes in the phase speed of individual waves subject to a fixed total baroclinic wave source. The phase speed of the wave source is included with the barotropic mechanism because it is fundamentally coupled to the changes in advection by the background U. Acting in isolation, the phase speed effect damps U anomalies (Lorenz (2014a); Lorenz (2014b)), so including the phase speed effect with the background flow weakens the barotropic feedback and strengthens the baroclinic feedback.

Let $S_0(c)$ be the distribution of the wave activity source in the control simulation as a function of phase speed, c. The portion of the new source coming from the baroclinic feedback is defined as S_0 scaled by the ratio of the new to control S amplitude:

$$\frac{\int_{-\infty}^{\infty} S_1(c)dc}{\int_{-\infty}^{\infty} S_0(c)dc} S_0(c), \tag{3}$$

where $S_1(c)$ is the wave activity source in the new simulation. The source for the phase speed change is defined as

$$\frac{\int_{-\infty}^{\infty} S_0(c)dc}{\int_{-\infty}^{\infty} S_1(c)dc} S_1(c), \tag{4}$$

which is the new source scaled to have the same integrated amplitude as the original source.

2) Baroclinic RWC

Here the barotropic RWC is extended to a linearized baroclinic model. Baroclinic RWC improves the simulation of the eddy momentum fluxes compared to the barotropic model, while also involving fewer heuristic assumptions. For example, the baroclinic source is not parameterized as an external force as in Barotropic RWC but is instead simulated directly by the dynamical equations. The linearized equations for the upper- and lower-level QG model are:

$$\begin{aligned} \frac{\partial q_1'}{\partial t} + \frac{\bar{u}_1}{a\cos\phi}\frac{\partial q_1'}{\partial \lambda} + \gamma v_1' - k_T k_D^2 \left(\psi_1' - \psi_2'\right) &= F_{q1}' + \nu \nabla^2 q_1', \\ \frac{\partial q_2'}{\partial t} + \frac{\bar{u}_2}{a\cos\phi}\frac{\partial q_2'}{\partial \lambda} + \gamma v_2' - k_T k_D^2 \left(\psi_2' - \psi_1'\right) &= F_{q2}' + \nu \nabla^2 q_2', \end{aligned} \tag{5}$$

where q is the Quasi-Geostrophic (QG) Potential Vorticity (PV) ($q_j = f + \nabla^2\psi_j - k_D^2\left(\psi_j - \psi_{3-j}\right)$ where ψ is the streamfunction), k_D is the inverse deformation radius, γ_j is the zonal-mean meridional PV gradient ($= a^{-1}\partial_\phi q_j$), k_T and k_M are the thermal and mechanical damping rates, level 1 is the upper level and level 2 is the lower level. The inverse deformation radius is given by:

$$k_D^2 = \frac{f^2}{\frac{1}{2}c_p\left(\left(\frac{p_2}{p_0}\right)^\kappa - \left(\frac{p_1}{p_0}\right)^\kappa\right)\left(\overline{\theta}_1 - \overline{\theta}_2\right)}, \tag{6}$$

where c_p is the specific heat at constant pressure, p is the pressure, $\kappa = 2/7$, $\overline{\theta}$ is the zonal-mean potential temperature from the atmospheric model or reanalysis data set, and the $\frac{1}{2}$ factor in the denominator makes the QG model consistent with the vertical differencing in the two-level primitive equation model. The values of k_T and k_M are taken from the GCM simulation we are attempting to model, however, for reanalysis or a comprehensive GCM, a linear approximation of the heating and damping must be used. In the baroclinic case, $F_{qj} + \nu\nabla^2 q_j$ parameterizes nonlinearities as forcing plus damping, which is like stochastic models of baroclinic turbulence (DelSole (2004)). Unlike most stochastic models, however, F_q does not have a simple *a priori* distribution (i.e. white noise), instead the forcing is calculated such that it gives a particular vertical EP flux. Like the barotropic RWC, we prescribe the vertical EP flux at each zonal wavenumber and phase speed so that it matches the GCM (for details, see Appendix). The diffusivity, ν, is the one free parameter. We use $\nu = 2.47 \times 10^6$ m^2/s, which minimizes the error in the eddy momentum flux integrated over phase speed and zonal wavenumber. This value of diffusivity is also sufficient to stabilize the system to baroclinic instability for all phase speeds and wavenumbers. The optimal ν depends on model resolution. At T85, for example, ν is almost 4 times smaller than the T42 value given above.

d. Linear Response Function (LRF) Analysis

Hassanzadeh and Kuang (2016) developed a novel method to force U perturbations with arbitrary latitude/pressure structure in a GCM. Using this method, the mean flow can be manipulated in various ways to better understand the dynamics (Hassanzadeh and Kuang (2019)). The methodology first involves running a large number of simulations, each with a zonally uniform prescribed forcing that varies in location (latitude and pressure) across the simulations. This exercise creates a linear operator (perturbations are assumed to be small enough) converting zonal-mean U and temperature forcing to U and temperature. Hassanzadeh and Kuang (2016) then implemented the method in Kuang (2010) to filter this ill-conditioned linear operator so that it can be inverted to find the forcing that creates an arbitrary U and temperature mean state.

We repeat this method in the two-level primitive equation model. All parameters and forcing profiles are the same as Hassanzadeh and Kuang (2016) except for the pressure dimension: instead of applying a Gaussian shaped forcing in pressure, we simply apply the forcing in either the upper

or lower level of the two-level primitive equation model. Our forcing experiments are run for 251000 days at T42 resolution (see section 2a2).

3. Results

a. Upper-Level Friction on Zonal-Mean Zonal Wind Anomalies

The time mean zonal-mean zonal wind (U) for the multi-level GCM with the control Held and Suarez (1994) parameters has a mid-latitude jet centered at about 42° latitude (Fig. 1a). Next, we calculate EOF1 of the instantaneous vertical- and zonal-mean zonal wind and then regress the zonal-mean wind on the resulting Principal Component (PC) (Fig. 1b). EOF1 has oppositely signed center of actions on either side of the jet maximum and therefore represents north/south shifts in the mid-latitude jet. In Fig. 1c, the lagged regression between U and $\overline{u'v'}$, and U and the EP fluxes (Edmon et al. (1980)) is averaged from 10 to 20 days *after* the peak in U. These eddy fields are lagged in time to isolate the component of the eddies that is caused by the U anomalies from the "random" eddy fluxes that generated the U anomalies in the first place (Lorenz and Hartmann (2001)). The $\overline{u'v'}$ anomalies "caused" by EOF1 are positive in the upper troposphere at latitudes between the two centers of action of EOF1. This leads to convergence of momentum at the positive center of EOF1 and divergence of momentum at the negative center, which reinforces the EOF1 U anomalies. The horizontal component of the EP fluxes is simply $\cos(\phi)$ times the $\overline{u'v'}$ itself; the vertical component of the EP fluxes are proportional to the eddy heat fluxes that generate the waves in the first place. In Fig. 1c, the vertical EP flux anomalies are larger than the horizontal EP fluxes, which previous studies have used to infer that anomalies in baroclinic wave generation are causing the $\overline{u'v'}$ anomalies. In the literature, this is called the baroclinic feedback mechanism.

In Fig. 2, the above diagnostics are quantified more precisely by calculating the EP flux budget of the layer from 500 to 100hPa.[1] First, define m to be the eddy momentum flux convergence ($-\frac{1}{a\cos^2\phi}\frac{\partial\cos^2\phi\overline{u'v'}}{\partial\phi}$) integrated from 500 to 100hPa and s to be the convergence of the vertical EP flux integrated from 500 to 100hPa divided by $a\cos\phi$. Note that

$$\frac{1}{a\cos\phi}\langle\nabla\cdot\mathbf{F}\rangle = -s+m, \tag{7}$$

[1]The GCM results below are not sensitive to changes in the upper troposphere layer boundaries. 500hPa is used as the boundary of the upper troposphere because most of the eddy momentum flux occurs above this level. This is not true of levels much higher than 500hPa.

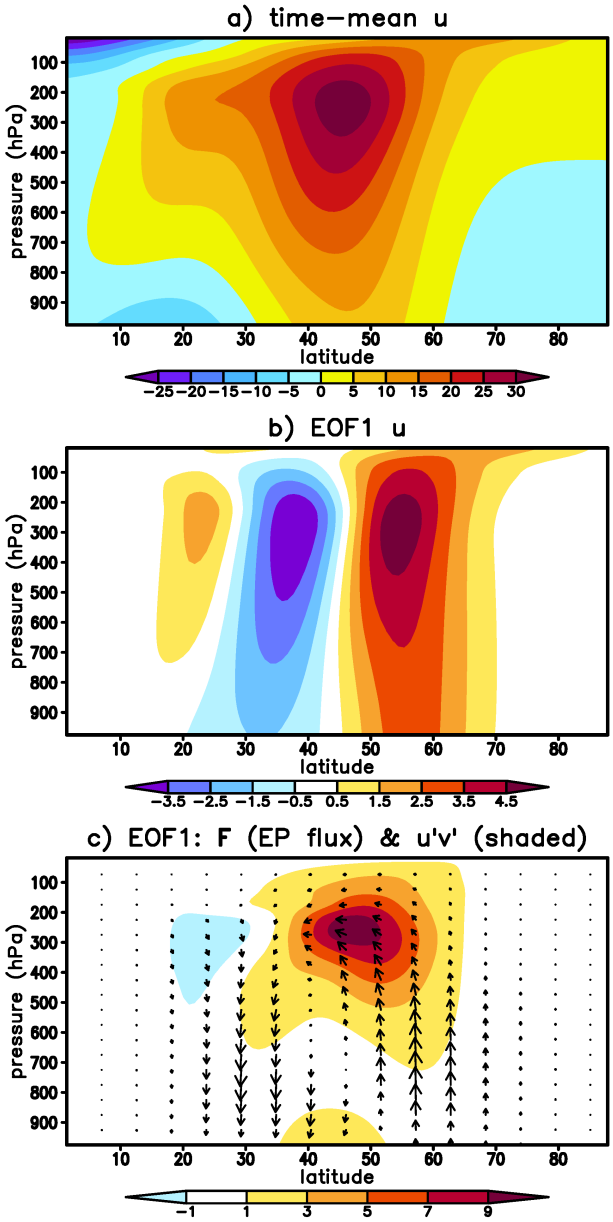

FIG. 1. a) Time- and zonal-mean zonal wind (U) from the control simulation of the multi-level GCM (m/s). b) U anomalies regressed on PC1 of instantaneous U variability (m/s). There is no time lag. c) $\overline{u'v'}$ (shaded, m^2/s^2) and EP flux (**F**, arrows) anomalies regressed on PC1 of instantaneous U variability. Results are averaged over lags 10 to 20 days. The ratio of the vertical to horizontal **F** arrow length is scaled so that zero **F** divergence looks like zero displayed-arrow divergence.

where **F** is the EP flux (Edmon et al. (1980)) and $\langle\cdot\rangle$ denotes a vertical average from 500 to 100hPa. Next, the lagged regression with the U PC is calculated. Finally, the m and s anomalies

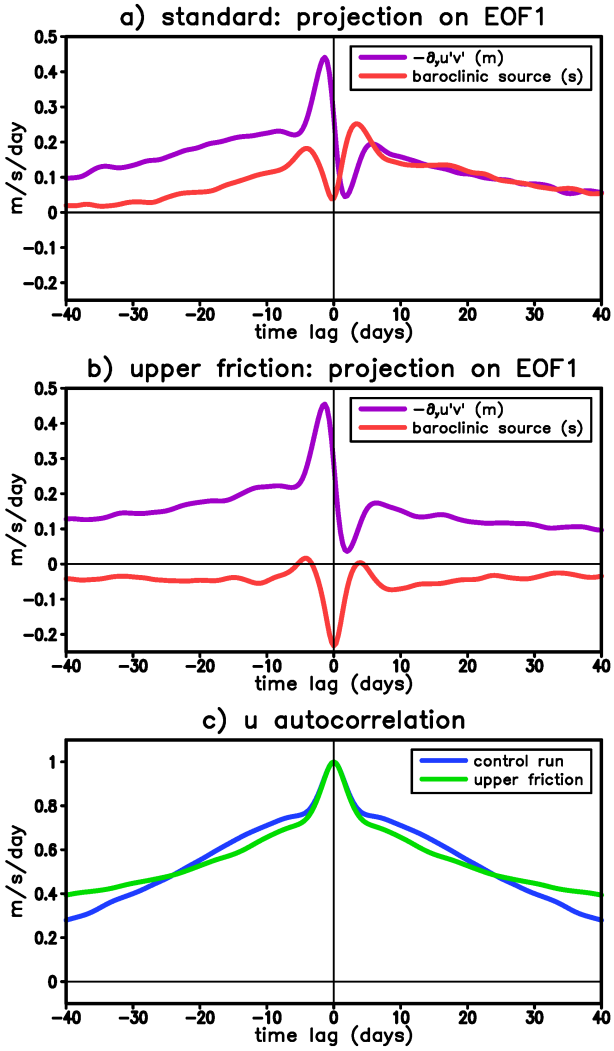

FIG. 2. Lagged regression of upper-troposphere wave activity budget on PC1 of U for the multi-level GCM. Positive lags mean that U leads in time. a) Eddy momentum flux convergence (purple) and baroclinic source (red, convergence of the vertical component of the EP flux (= $\mathbf{F}_p$) in the upper troposphere, see text). b) Same as (a) but for the experiment where friction acts at upper-levels for U anomalies (see text). c) Autocorrelation for PC1 of U for the control (blue) and upper friction (green) simulation.

are projected onto the U EOF1 pattern. The m projection (purple line, Fig. 2a) peaks before the U anomalies (negative lags), which is consistent with the fact that m forces U. The focus here is on the reinforcing m at positive lags beyond 7 days, which is consistent with a positive feedback (Lorenz and Hartmann (2001)). The baroclinic wave activity source, s, anomalies (red) attain the same values as m at positive lags beyond 7 days, which, consistent with the previous figure, suggests

that the baroclinic feedback is responsible for the $\overline{u'v'}$ anomalies. In the remainder of this paper, however, we show that the baroclinic feedback is not the main driver of $\overline{u'v'}$ anomalies. The issue, as described in more detail below, is that reasoning based on the anomalous EP flux budget alone is incorrect. Instead the relative sizes of the vertical and horizontal EP fluxes in the mean state must also be taken into account (see section 3d). First, we describe an experiment which demonstrates that s is not primarily responsible for m.

In addition to $\overline{u'v'}$ reinforcing U anomalies, the baroclinic feedback also requires the eddy fluxes to maintain temperature gradient anomalies that are collocated with U anomalies so that baroclinic wave source anomalies are also collocated with U. Robinson (2000) describes in detail how the shape of anomalous upper-level EP flux divergence acts to maintain such temperature gradient anomalies. This theory requires friction at low levels only. Therefore, if the Held and Suarez (1994) friction is moved to upper levels then the U anomalies are no longer associated with temperature gradient anomalies, which will presumably turn off the baroclinic mechanism. Completely changing the friction from lower to upper levels will dramatically change the mean circulation making the results hard to interpret. Therefore, we only change friction acting on U anomalies (recall U = zonal-mean zonal wind). Specifically, the zonal wind equation in the upper-friction model is:

$$\frac{\partial u}{\partial t} + \text{adiabatic terms} = -k_M\left(u' + \overline{\overline{u}}\right) - k_U\left(\overline{u} - \overline{\overline{u}}\right), \tag{8}$$

where the friction is on the right-hand side, the double bar, $\overline{\overline{u}}$, means a time- and zonal-mean, k_M is the standard low-level Held and Suarez (1994) friction profile and k_U is an alternative friction profile acting at upper levels. The meridional wind, temperature and surface pressure equations are unchanged relative to the control simulation. Note that in (8), k_U acts on zonal-mean zonal wind anomalies only, the remaining portion of the zonal wind feels the standard friction. Also, note that if $k_M = k_U$, the friction in (8) collapses to the standard form: $-k_M u$. Here the k_U profile is given by

$$k_U = \begin{cases} 0.37\text{day}^{-1}\cos^2\left(\pi\frac{p-300hPa}{600hPa}\right), & \text{if} p < 600hPa \\ 0, & \text{otherwise,} \end{cases} \tag{9}$$

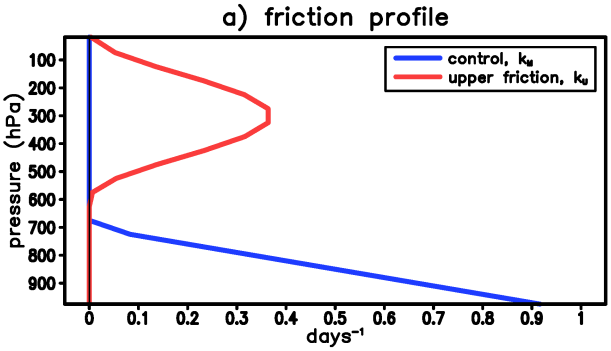

FIG. 3. Rayleigh Friction damping profile in the multi-level GCM for standard Held and Suarez (1994) forcing (blue). This profile also acts on the eddies and time-mean U profile in the upper-friction simulation. The Rayleigh Friction profile acting on U anomalies only for the upper-level friction simulation is shown in red.

where the factor 0.37 gives a vertically-integrated friction time scale (calculated as in Lorenz and Hartmann (2001)) comparable to the control simulation. For the results below, the details of the friction profile are not important as long as the friction is concentrated at upper levels. The friction profiles are plotted in Fig. 3. The changes in forcing in (8) are relatively small compared to previous studies that have manipulated U EOF profiles to understand the dynamics (Zurita-Gotor et al. (2014); Hassanzadeh and Kuang (2019)). For example, the friction profile, (9), is no larger than the standard friction and, moreover, only needs to be applied to a relatively thin layer of the atmosphere. Previous studies have used very strong mechanical and/or thermal forcing at all levels to accomplish an alternate EOF profile (Zurita-Gotor et al. (2014); Hassanzadeh and Kuang (2019)). The eddy heat flux response in these strongly forced simulations is very large and non-intuitive. For example, the baroclinic source in Zurita-Gotor et al. (2014) and Hassanzadeh and Kuang (2019) strongly opposes barotropic U EOFs leading to a negative baroclinic feedback. For our experiment, on the other hand, the eddy heat flux response is weak and therefore the separation of baroclinic versus barotropic processes in cleaner. We explore the negative baroclinic feedbacks seen in Zurita-Gotor et al. (2014) and Hassanzadeh and Kuang (2019) below.

In Fig. 4, the results in Fig. 1 are repeated for the upper friction on U anomalies simulation. The time-mean U is practically unchanged from before. The leading EOF has very similar structure in latitude, but the vertical structure is much more barotropic, although not completely so (Fig. 4b). The $\overline{u'v'}$ anomalies (Fig. 4c) are the same amplitude as the control simulation, but the vertical EP fluxes in the new simulation have disappeared. Evidently, the $\overline{u'v'}$ anomalies do not require

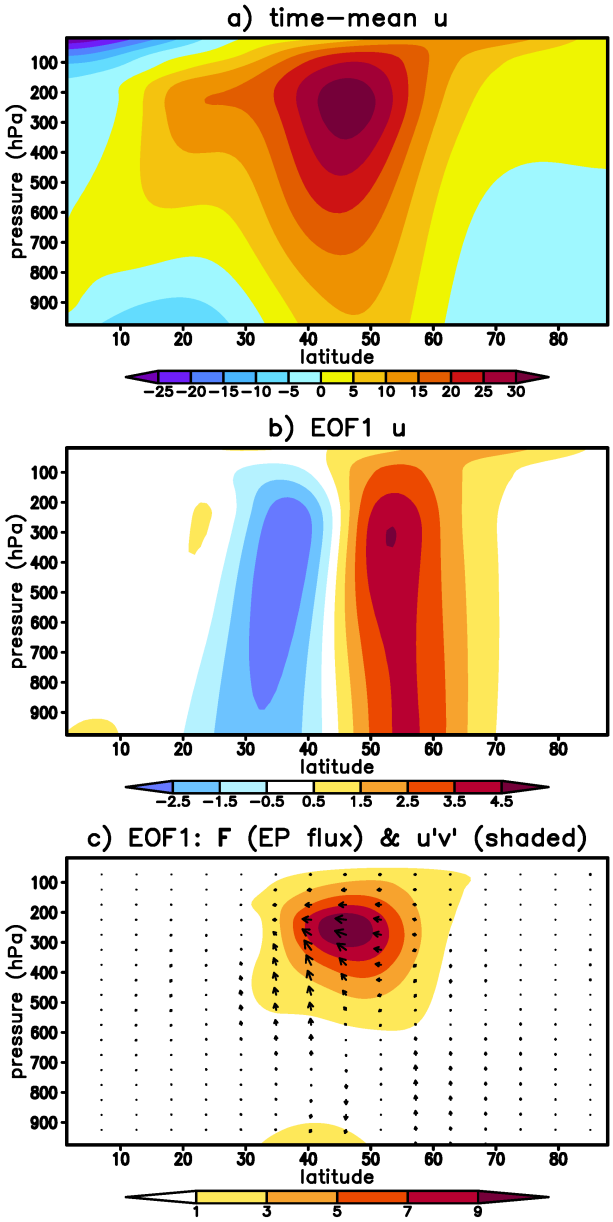

FIG. 4. Same as Fig. 1 but for the simulation with upper-level friction on U anomalies.

baroclinic source anomalies to maintain themselves. In fact, when quantifying the baroclinic source averaged over 500 - 100hPa (s, Fig. 2b), the s projection on EOF1 is actually slightly negative. The m projection is essentially unchanged although the persistence of the m anomalies in time is actually enhanced. The results of the lagged correlation are confirmed by the autocorrelation of the U PC (Fig. 2c), which shows that the persistence of U is relatively unchanged in the upper-friction experiment. However, there are subtle differences in autocorrelation that pertain to baroclinic

feedbacks and the poleward propagation of U anomalies, which is the subject of future research. The upper-friction experiment shows that m anomalies can be maintained without s anomalies. This does not mean that the vertical EP flux anomalies are not important for the zonal-mean momentum budget and the vertical structure of EOF1 in Fig. 1b, only that the vertical EP flux anomalies are not the direct cause of the majority of the m anomalies. Below, we will return to a similar upper friction experiment in the two-level model to make quantitative estimates of the baroclinic versus barotropic feedback.

b. Rossby Wave Chromatography (RWC)

The second line of evidence suggesting the baroclinic feedback is not dominant comes from RWC experiments. In RWC, the baroclinic wave source and background flow are prescribed independently in a linearized model and therefore the role of each can be separately quantified. As discussed above (Section 2c), the change in phase speed of individual waves subject to a fixed total baroclinic wave source is included in the background-flow/barotropic mechanism. Because the phase speed effect damps most U anomalies (Lorenz (2014a); Lorenz (2014b)), including the phase speed effect in the barotropic mechanism actually strengthens the argument that the baroclinic mechanism is not as important.

Because EOF's are by definition transient, we force a permanent EOF1 state in the two-level GCM using the LRF methodology of Hassanzadeh and Kuang (2016) (see section 2d). Unlike a fluctuating EOF1, a steady EOF1 state enables the phase speed spectra to be easily calculated. The upper-level $\overline{u'v'}$ phase speed spectrum for the positive phase of EOF1 in the GCM is shown in Fig. 5a. $\overline{u'v'}$ increases at high phase speeds at and poleward of the mean jet (centered at $\approx 42°$). There are two zones of weakening $\overline{u'v'}$: one at low phase speeds centered on the mean jet and one in the subtropics at moderate phase speeds. This response looks similar to the response to decreases in the zonal-mean component of friction in Lorenz (2015) (see their Fig. 5a) except the second zone of decreasing $\overline{u'v'}$ is missing.

The barotropic RWC (Fig. 5b) gets the general pattern but (1) the increases in $\overline{u'v'}$ are too confined in latitude, (2) there are spurious decreases on the poleward side of the increases, (3) the decreases in the subtropics are too strong and (4) the decreases at low phase speed are too weak. The first two biases are also present in the response to decreases in the zonal-mean component

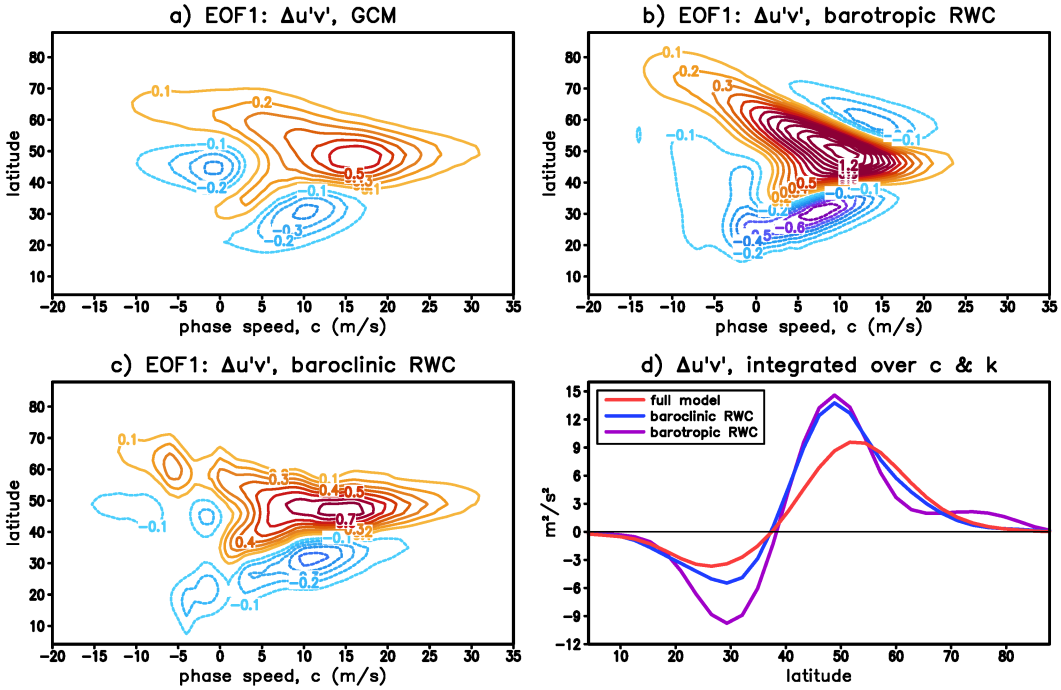

FIG. 5. Comparison of the $\overline{u'v'}$ response to EOF1 in the two-level GCM and two versions of RWC applied to the two-level GCM. a) GCM phase-speed/latitude spectrum integrated over zonal wavenumber. b) Barotropic RWC phase-speed/latitude spectrum integrated over zonal wavenumber. c) Baroclinic RWC phase-speed/latitude spectrum integrated over zonal wavenumber. d) The response of the meridional convergence of $\overline{u'v'}$ integrated over phase speed: GCM (red), barotropic RWC (purple), baroclinic RWC (blue).

of friction seen in Lorenz (2015). The baroclinic RWC (Fig. 5c) improves the simulation of the regions of decreasing $\overline{u'v'}$. In particular, the spurious region of decreasing $\overline{u'v'}$ on the poleward side is no longer present. The distribution of the $\overline{u'v'}$ increases are also more evenly distributed in latitude and phase speed, which is consistent with the GCM. Integrating the $\overline{u'v'}$ response over phase speed (Fig. 5d), we see that the baroclinic RWC improves the $\overline{u'v'}$ response at all latitudes. The main error in the baroclinic RWC is the overestimation of the response around 50°. This bias is shared with the barotropic model.

In Fig. 6, the baroclinic RWC is used to isolate mechanisms for the $\overline{u'v'}$ anomalies associated with EOF1 of U. The baroclinic feedback (Fig. 6a) explains a significant portion of the $\overline{u'v'}$ decreases in the subtropics at moderate phase speeds. The changes in baroclinic wave source that are driving the baroclinic feedback (Fig. 6b) are largest in the subtropics, which is consistent with Fig. 6a. However, there are also significant increases in source poleward of 50°. Apparently

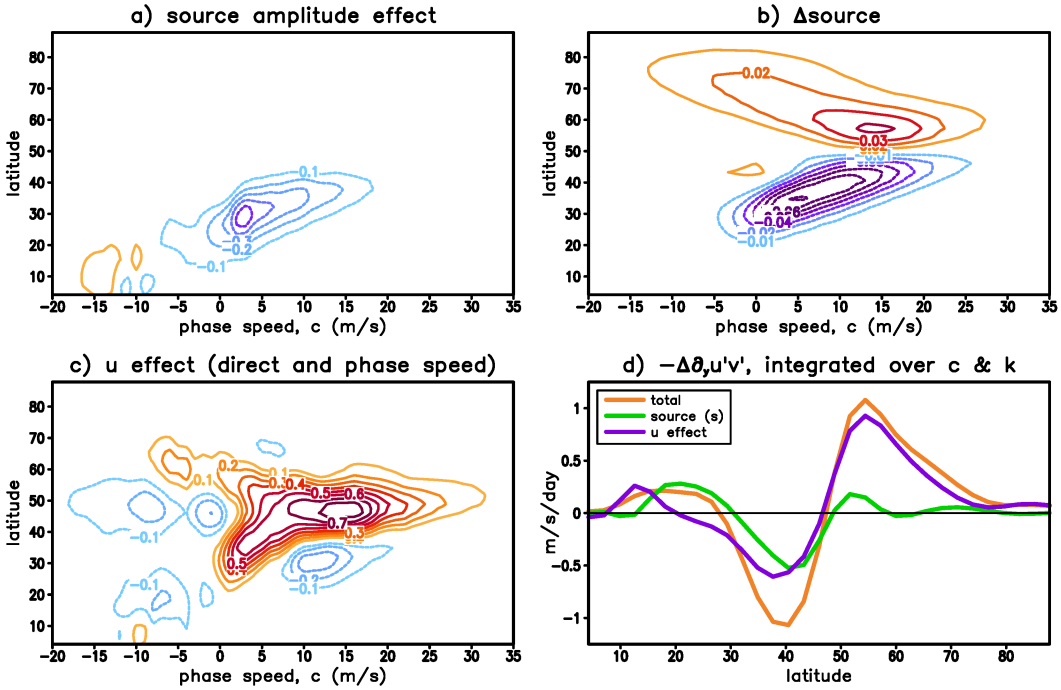

FIG. 6. Phase-speed/latitude spectrum of the response to EOF1 partitioned into various mechanisms using baroclinic RWC applied to the two-level GCM (see text). a) Change in $\overline{u'v'}$ due to baroclinic wave source amplitude. b) For reference, the change in baroclinic wave source. c) Change in $\overline{u'v'}$ due to background flow and baroclinic source phase speed effect. d) $\overline{u'v'}$ response in previous panels integrated over phase speed: total response from baroclinic RWC (orange), baroclinic source amplitude effect (green), background flow and baroclinic source phase speed effect (purple).

these sources are in regions of phase speed/latitude space that are unfavorable for Rossby wave propagation. The barotropic mechanism (background flow and phase speed changes, Fig. 6c) accounts for all the increases in $\overline{u'v'}$ and a portion of the $\overline{u'v'}$ decreases. Fig. 6d shows the eddy momentum flux convergence, m, integrated over phase speed. We see that the m decreases seen equatorward of 45° are about equally split between barotropic and baroclinic mechanisms while the m increases further poleward are almost exclusively due to barotropic mechanisms. To get a single number quantifying the relative contribution of each mechanism on the total, we project each mechanism's m onto the total m and normalize by the total m amplitude ($= \sqrt{\sum_\phi m^2}$). According to this metric, the barotropic feedback contributes 70.3% to the eddy feedback and the baroclinic feedback contributes 28.1% (see first row of Table 1). We repeated this analysis but for barotropic RWC and the contribution of the barotropic feedback is even larger (Table 1). However, because

TABLE 1. Estimates of the relative contribution of barotropic and baroclinic mechanisms to the eddy feedback in the two-level GCM using different methods (see text).

Method	Experiment	Barotropic Feedback (%)	Baroclinic Feedback (%)
RWC	Baroclinic RWC	70.3	28.1
	Barotropic RWC	91.2	8.1
LRF EOF	Barotropic	62.8	37.2
	Baroclinic	71.5	28.5
	Strong Thermal Damping	77.0	23.0
	Upper Friction	87.1	12.9
	Force U only	70.1	29.9
IWP model		68.8	30.5

the baroclinic RWC gives more realistic eddy fluxes, the baroclinic RWC results should be given more weight.

c. Linear Response Function (LRF) Analysis

Hassanzadeh and Kuang (2016) developed a novel LRF methodology, which can be used to force arbitrary time- and zonal-mean U profiles in a GCM. For example, Hassanzadeh and Kuang (2019) used this methodology to impose a barotropic version of EOF1 in a GCM. In this paper, we implement the LRF methodology in the two level model and we manipulate the U EOF1 pattern in order to understand the dynamics of the eddy feedback. In some cases, the manipulated EOFs lead to strongly negative baroclinic feedbacks, which offset the positive barotropic feedback. This diversity of baroclinic feedback strength allows us to isolate its relative contribution. For our analysis below, we assume *both* feedback mechanisms are operating. We can disentangle their relative contributions by comparing the baroclinic source and eddy momentum flux response to *two* different EOF1 structures (i.e. standard and barotropic). We do this for a variety of different EOF1 structures and find similar estimates for the relative contribution of baroclinic and barotropic feedbacks for the standard EOF1 in each.

First, we look at the LRF forced version of EOF1: positive phase minus negative phase (Fig. 7a). The upper and lower U (blue) are in phase but the lower U is smaller in amplitude, which implies the temperature gradient is collocated with the U anomalies. The eddy momentum flux convergence at the upper level (m, purple) is mostly in phase with U, consistent with the positive

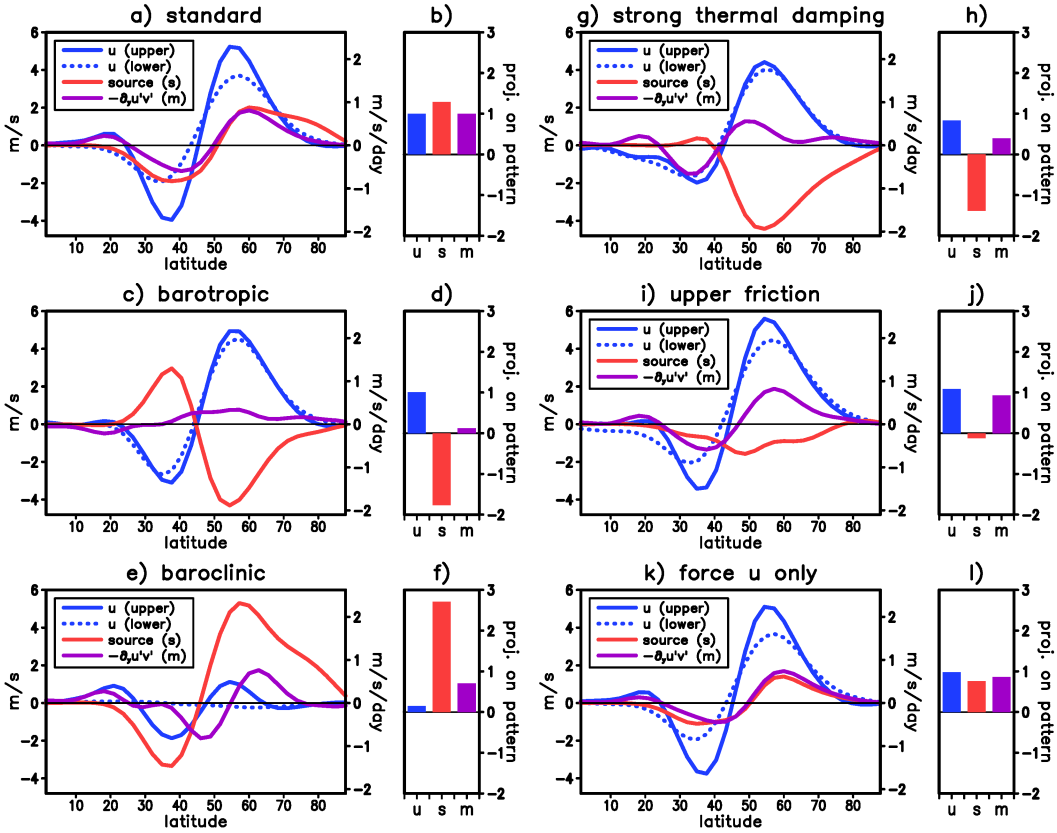

FIG. 7. Results of LRF experiments with prescribed EOF1 structures in the two-level GCM. a) Standard EOF1 U (blue, solid = upper-level, dashed = lower level). Response to standard EOF1 for baroclinic source (red) and eddy momentum flux convergence (purple). b) Blue: projection of U response onto the standard EOF1 (identically 1 in this case). Red: projection of baroclinic source response onto the standard eddy momentum flux convergence pattern. Purple: projection of eddy momentum flux convergence response onto the standard eddy momentum flux convergence pattern (identically 1 in this case). c) Same as (a) but for barotropic EOF1. d) Same as (b) but for barotropic EOF1 projected on standard patterns. e) and f) Same as (a) and (d) but for purely baroclinic EOF1. g) and h) Same as (a) and (d) but for strong thermal damping EOF1. i) and j) Same as (a) and (d) but for upper-level friction on U anomalies EOF1. k) and l) Same as (a) and (d) but for EOF1 forced via U instead of U and temperature.

feedback, although m is displaced slightly poleward compared to U. The convergence of vertical EP flux in the upper level, or baroclinic wave source, (s, red) is aligned closely with m but of slightly larger amplitude. The fact that s is larger than m everywhere is consistent with the baroclinic feedback, although we will see that this is deceptive. For diagnosing the feedback mechanisms,

we would like to quantify the U, m and s response with a single number. For U, we project the U (dependent on both latitude and pressure) on the standard EOF1 U and normalize by the standard U amplitude. By definition, this metric is one for the standard EOF (blue bar in Fig. 7b). Because s drives m according to the baroclinic feedback, s (dependent on latitude only) is projected on the standard EOF1 m and normalized by the standard m. Because s is projected on m, the amplitude is different than one (= 1.29) in the standard EOF case (red bar in Fig. 7b). Like U, which is projected on itself, m (dependent on latitude only) is projected on the standard m (purple bar in Fig. 7b).

For the next experiment, the target U anomaly is barotropic with U equal to the vertical average of the standard U. The resulting U is slightly off from completely barotropic (blue, Fig. 7c) but still very close. Consistent with Hassanzadeh and Kuang (2019), the m response disappears for barotropic EOF1 anomalies. Hassanzadeh and Kuang (2019) argued that this result is evidence that the baroclinic feedback is dominant. However, barotropic U do not give zero s, instead s is strongly opposite U, implying a *negative* baroclinic feedback in this experiment. Currently, we do not understand *why* s is opposite U.[2] But if one accepts that *both* baroclinic and barotropic feedbacks are operating, then this experiment provides a way to disentangle their relative contributions (both for standard EOF and the barotropic EOF). For example, the s projection decreases from 1.29 in Fig. 7b to -1.77 in Fig. 7d (note: the units are not important for this discussion of relative changes). Meanwhile, m decreases from 1.0 to 0.12 and U hardly changes. Therefore, a 3.06 decrease in s leads to a 0.88 decrease in m, which implies that 1 unit of s leads to 0.29 (= 0.88/3.06) units of m. Because the s amplitude in the standard EOF case is 1.29, the fraction of m due to the baroclinic feedback in the standard case is 0.37 (= $1.29 \cdot 0.29$). More generally, with two simulations, we can find the two coefficients that determine the strength of the barotropic and baroclinic feedback:

$$\begin{aligned} a_U U_0 + a_s s_0 &= m_0, \\ a_U U_1 + a_s s_1 &= m_1, \end{aligned} \tag{10}$$

where U, s and m are the projection values in the bar chart in Fig. 7, the subscripts 0 and 1 refer to the standard EOF1 and alternate EOF1, respectively, and a_U and a_s are constant coefficients

[2]James (1987) and Nakamura (1993) show how barotropic shear can suppress baroclinic instability. In our case, however, the changes in baroclinic source are collocated with the U changes, which means the baroclinic instability changes are strongest where barotropic shear changes are weakest. Therefore, we believe an alternative explanation is required.

that are calculated from the simple system of two equations. After the coefficients are determined, $a_U U_0/m_0$ and $a_s s_0/m_0$ are the relative contribution of the barotropic and baroclinic feedback for the standard EOF1 (note: these quantities are dimensionless and represent the fractional contribution). Similar expressions for the non-standard EOF structure involve subscripts of 1 instead of 0. The estimates for the standard EOF, which are given in Table 1, agree with the simple estimate in this particular case. Although Hassanzadeh and Kuang (2019) did not analyze the EP flux budget for their barotropic EOF, their heat fluxes were the opposite sign as U, which is consistent with our results.

To test the robustness of the above estimate, we can also impose alternative U anomalies. In Fig. 7ef, the baroclinic portion of EOF1 is imposed: the lower level U is zero and the upper level U is the difference between upper and lower in the standard case. In this baroclinic EOF case, the barotropic feedback is small (blue bar in Fig. 7f), so almost all the feedback is baroclinic. Apparently, for an almost pure baroclinic feedback, s leads to m anomalies that are about four times smaller (compare red and purple bars in Fig. 7d), which suggests that s must be several times greater than m for baroclinic feedbacks to be the sole mechanism. This contrasts with previous studies that assume a one-to-one ratio of s to m is sufficient for the baroclinic feedback to be dominant. An understanding of the large ratio of s to m is provided in Section 3d. Applying the quantitative model in (10), the estimates for the barotropic and baroclinic feedback for the standard EOF from this case are 71.5% and 28.5%, respectively, which is comparable to the previous case.

An alternate way to generate a barotropic EOF is to strongly damp zonal-mean temperature anomalies back to climatology. In contrast to imposing a barotropic version of EOF1, the eddies have some control over the shape of the EOF. This experiment is inspired by similar experiments in Zurita-Gotor et al. (2014). For this experiment, the damping time scale is set to 0.1 days^{-1} and the leading EOF1 structure was calculated from the model simulation. Next, we imposed this alternate EOF structure in the control simulation using the LRF methodology (Fig. 7g). The U response is nearly barotropic like the previous barotropic experiment (Fig. 7c), however, the equatorward center of action extends to lower latitudes in the strong thermal damping experiment. The U, s and m projections in the new experiment are similar to the barotropic experiment and the estimates for the barotropic and baroclinic feedback for the standard EOF from this case are 77.0% and 23.0%, respectively (Table 1). Like the barotropic case, we believe the correct interpretation

of the strong thermal damping case depends critically on recognizing the large negative baroclinic source feedback that accompanies barotropic U anomalies. Assuming the s response is negligible when U is barotropic leads to the erroneous conclusion that the barotropic feedback is negligible (Zurita-Gotor et al. (2014); Hassanzadeh and Kuang (2019)).

In Fig. 7i, we repeat the upper friction experiment from Section 3a in the two level model, and then prescribe EOF1 from this simulation using the LRF methodology. As in the multi-level model, the s response in this simulation is small and therefore the estimate of baroclinic feedback is especially small for this case (12.9%). We do not understand why the estimate is lower in this case but it is interesting that the two cases where the EOF structure is determined by the GCM (Upper Friction and Strong Thermal Damping) have smaller baroclinic feedback estimates than the two cases where the EOF structure is simply prescribed (Barotropic and Baroclinic).

Most of the above experiments involve dramatic changes in EOF1 structure. In the final experiment, we try a subtle change in EOF structure. Hassanzadeh and Kuang (2016) develop the LRF matrix by forcing both U and temperature. All the experiments above follow this convention. In the last experiment, we develop a separate LRF matrix that involves the forcing of U only. The forced EOF1 structure in the experiment (Fig. 7k) is very close to the standard experiment that forces both U and temperature (Fig. 7a). The differences between the experiments are more evident in the relative amplitude of s and m. Using these two experiments to constrain the two feedback mechanisms (10) gives 70.1% and 29.9% for the barotropic and baroclinic feedbacks, respectively. This estimate is the median of all LRF experiments and is very close to the Baroclinic RWC estimate. It is possible that the estimates from this experiment should carry more weight because the EOF structure is not much different from the control, however this is just conjecture.

d. Inferred Wave Propagation (IWP) Model Based on Mean State

For our final estimates, we use the meridional wave propagation implied by the time-mean EP flux budget to create a model that predicts the anomalous meridional wave activity flux (= negative of the eddy momentum flux) resulting from an anomalous wave source. We call this simple model the Inferred Wave Propagation (IWP) model. IWP makes the ideas of Robert et al. (2017) on radiation efficiency quantitatively precise. The model is motivated by the upper-level wave activity

budget (Edmon et al. (1980)):

$$\frac{\partial A}{\partial t} + \nabla \cdot \mathbf{F} = -D, \tag{11}$$

where A is the wave activity, $\mathbf{F}$ is the EP flux vector (ϕ and p components), D is the dissipation (typically $D > 0$), and for simplicity we use cartesian coordinates initially. Noting that the divergence of the vertical component of $\mathbf{F}$ is the negative of the baroclinic wave source and considering the steady state solutions only:

$$\frac{\partial F}{\partial \phi} = s - D, \tag{12}$$

where F is the meridional component of the EP flux and ϕ is the latitude. For northward propagating waves, it physically makes the most sense to integrate (12) from south to north to find F. As (12) is integrated, F increases via the baroclinic wave source ($s > 0$) and decreases due to dissipation. We write the dissipation as

$$D = \frac{F}{L}, \tag{13}$$

where L is the e-folding spatial scale characterizing the amount of decay of F with distance from a point source. Equation (13) is technically not an approximation because L is allowed to vary in latitude. L determines the meridional propagation. For example, if L is large, then the decay of wave activity is weak and therefore waves propagate large distances in latitude away from their source. If L is small, then waves decay rapidly with latitude and meridional wave propagation is weak. One expects L to decrease as one approaches a critical latitude to account for the fact that a critical latitude damps wave activity and therefore limits meridional wave propagation beyond it. Because L determines the meridional wave propagation, changes in L represent the barotropic mechanism. On the other hand, changes in s while keeping L constant represent the baroclinic mechanism.

We have only considered waves propagating in a single direction. In the IWP model, a certain fraction of the waves propgate northward, w_n, and the remaining waves propagate southward, $w_s = 1 - w_n$. Therefore, (12) and (13) become:

$$\begin{aligned} \frac{dF_n}{d\phi} &= w_n s - \frac{F_n}{L}, \\ \frac{dF_s}{d\phi} &= w_s s + \frac{F_s}{L}, \end{aligned} \tag{14}$$

where the subscripts n and s refer to northward and southward propagating waves. As stated previously, the northward equation is integrated from south to north starting with $F_n = 0$ at the south-pole and the southward equation is integrated from north to south starting with $F_s = 0$. Due to the direction of integration in (14), $F_n \geq 0 \geq F_s$. The boundary conditions implicitly assume no wave reflection at the poles. The total wave activity flux, F, is

$$F = F_s + F_n. \tag{15}$$

IWP first solves (14) in the *inverse* sense: L and w are calculated from the time-mean s and F. Both L and w are assumed to vary with latitude. Once the meridional propagation characteristics of the time-mean flow (i.e. L and w) are known, the model can be applied to wave source anomalies to understand their impact on F anomalies (= negative of the eddy momentum flux). This quantifies the baroclinic feedback. While it is true that (14) is not a sophisticated model of wave propagation, it captures the role of dissipation in weakening the response of F to source anomalies, which is ignored in studies that simply compare the amplitude of terms in the *anomalous* wave activity budget (e.g. Lorenz and Hartmann (2001), Zurita-Gotor et al. (2014), Nie et al. (2014)). These studies ignore the efficiency of wave propagation out of the jet as inferred from the *mean* wave activity budget (Robert et al. (2017)). For example, in the mean state of the two-level GCM, the baroclinic wave source is significantly larger than the divergence of F in the mean state (note that the red line in Fig. 8a is closer to the blue than to zero in the jet core, $45°$), which means most wave activity dissipates locally rather than propagating meridionally. This implies that an anomalous baroclinic wave source must also be significantly larger than the divergence of anomalous F for the baroclinic feedback to be dominant (Robert et al. (2017)).

When solving (14) in the inverse sense, there are two knowns (F and s) but three unknowns (L, F_s and w_s, note the n subscripts can be found from the s subscripts). Therefore the problem is under-constrained. To make headway, we assume a simple case where the wave-propagation-direction is determined by the sign of F:

$$\begin{aligned} &w_n = 1,\ w_s = 0,\ \text{if}\ F > 0 \\ &w_n = 0,\ w_s = 1,\ \text{otherwise.} \end{aligned} \tag{16}$$

In future work we will explore the sensitivity to alternative w_s profiles. Consider a latitude zone where $F > 0$ and $w_n = 1$. If we use the approximation $F_n = F$ within this zone, then the F_n equation becomes:

$$\frac{dF}{d\phi} = s - \frac{F}{L}. \tag{17}$$

The divergence of the EP flux, E, is by definition $-s + \frac{dF}{d\phi}$. Therefore a simple manipulation of (17) implies that the length scale is given by: $L = -\frac{F}{E}$. Note that L is positive because E is negative. When $F < 0$ and $w_s = 1$, the same manipulation of terms gives $L = +\frac{F}{E}$. Generalizing for both cases,

$$L = \left|\frac{F}{E}\right|. \tag{18}$$

Once L, w_n and w_s are known, the wave activity flux (and eddy momentum flux) can be found by integrating each component of (14) over latitude starting at the appropriate pole. At some latitudes, L can be quite small, which causes standard explicit schemes to be unstable when integrating on the GCM latitude grid. Since (14) is linear, the easiest approach is to use absolutely stable implicit schemes. Here we use the 4th order Backward Differentiation Formula (BDF) (Curtiss and Hirschfelder (1952); Süli and Mayers (2003)). For initializing, we incrementally use with the 1st, 2nd and 3rd order BDF schemes.

To apply (14) to the spherical geometry, F, s and E should be scaled by $\cos\phi$ as follows:

$$\begin{aligned} F &= -\overline{u'v'}\cos^2\phi \\ s &= -\cos\phi \int_{500\text{hPa}}^{100\text{hPa}} \frac{\partial F_p}{\partial p}\partial p \\ E &= \cos\phi \nabla \cdot \mathbf{F}, \end{aligned} \tag{19}$$

where $\mathbf{F}$ is the two dimensional, spherical-geometry EP flux vector given in Edmon et al. (1980), F_p is the vertical EP flux vector in Edmon et al. (1980) and the integral in the s equation is over the upper troposphere.

The IWP model is first applied to the upper level of the two-level primitive equation model. Because the model is trained on the mean state, the time-mean EP flux divergence is very well captured (Fig. 8a, compare red line to dashed purple line). This result simply validates the assumption that $F = F_n$ when $w_n = 1$ and $F = F_s$ when $w_s = 1$, which is used in the derivation of L

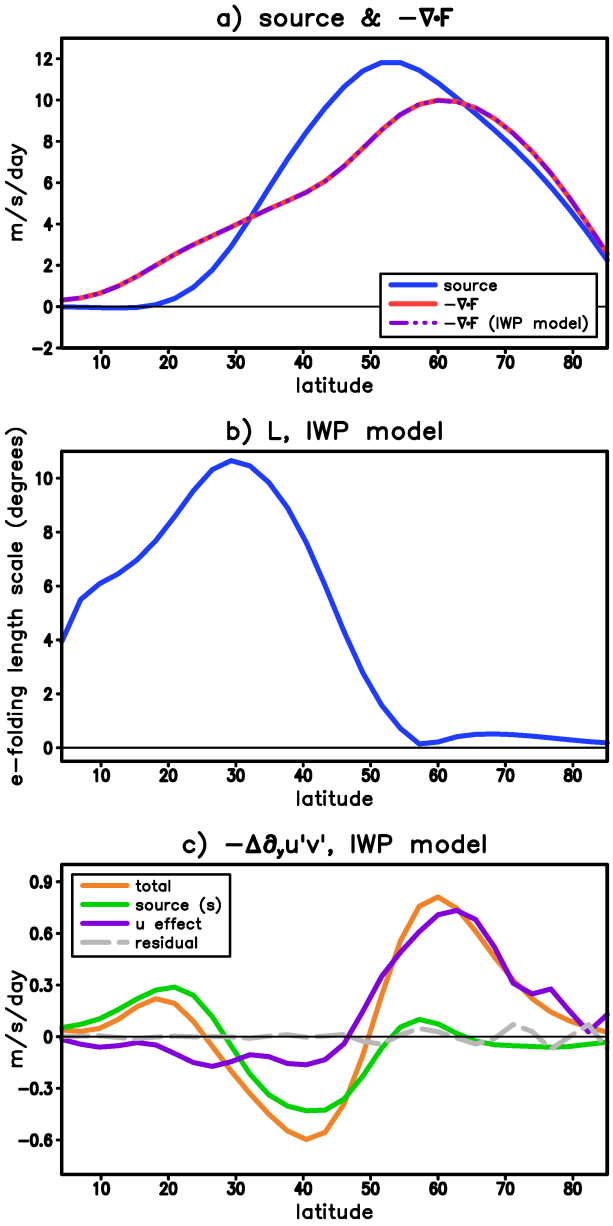

FIG. 8. a) Time-mean baroclinic source (blue) and EP flux convergence (red) at the upper-level of the two-level GCM. EP flux convergence from IWP model (purple dashed). b) The e-folding length scale, L, in the IWP model (degrees). c) Mechanisms of eddy momentum flux response to EOF1 from IWP model: total (orange), baroclinic source (green), meridional propagation or barotropic (purple) and residual (gray), which is the total minus the baroclinic and barotropic.

in (18). For example, near the boundaries where F changes sign, F is not strictly F_n or F_s because F from the nearby "zone" propagates across the boundary. Fig. 8a shows that this effect is small in

this case. L (Fig. 8b) maximizes in the subtropics around 30°, which is the same latitude that the wave source, s, and the negative EP flux divergence, $-E$, cross in Fig. 8a. This is consistent with the fact that a strong wave activity flux is required here to transport from latitudes where $s > -E$ to latitudes where $s < -E$. In the polar regions, on the other hand, the e-folding scale (L) is very small, which is consistent with the fact that little wave propagation is needed because $s \approx -E$

In Fig. 8c, we show the change in eddy momentum flux convergence, m, in response to the EOF1 from the LRF simulations (orange line). To estimate the baroclinic feedback, the change in s is input to the IWP model to estimate m in the absense of changes in wave propagation (i.e. L, w_n and w_s are unchanged). This m from the baroclinic feedback (green) is remarkably similar to the baroclinic feedback estimated from RWC (Fig. 6d). In both cases, the baroclinic feedback is very weak for the 60° center of action, maximizes at the 40° center of action and dominates the changes deep in the subtropics. For the barotropic feedback, we first calculate L, w_n and w_s in the new simulation. Next, the eddy momentum flux response to the original baroclinic wave source is calculated from (14). Finally, the momentum flux response from the control IWP is subtracted to get the change from barotropic processes (i.e. meridional propagation). The barotropic feedback (purple line) dominates the m response for the 60° center of action. As before, the relative contribution of each mechanism to the total m is found by projecting each mechanism's m onto the total m and normalizing by the total m amplitude ($= \sqrt{\sum_\phi m^2}$). In this case, the relative contribution of the barotropic and baroclinic mechanisms to the total m response is 68.8% and 30.5%, respectively (Table 1). This agrees with our previous estimates. Note that the sum of the two estimates does not add to 100% because there exists a second order term involving the change in source and L, w together.

The IWP model can also be compared with the LRF results for the baroclinic feedback for the non-standard EOF1 structures. For example, the baroclinic feedback for the strong thermal damping EOF1 is large and negative (Fig. 7gh). In Fig. 9, we show the amplitude of the baroclinic feedback for the five non-standard EOF structures in Fig. 7 for the IWP model versus the LRF estimates (see section 3c). The close correspondence suggests that the IWP model is in fact able to separate the contributions of the baroclinic and barotropic feedback.

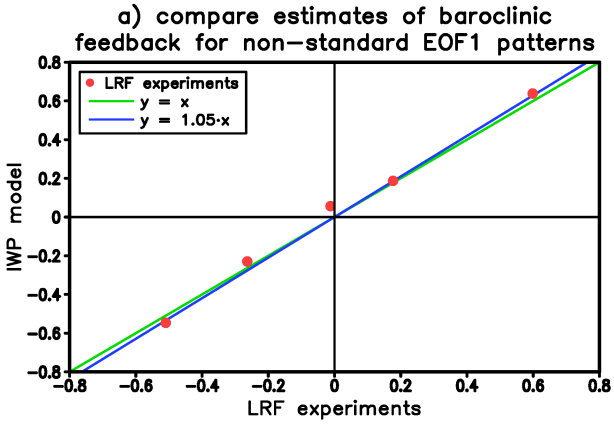

FIG. 9. For the two-level GCM, scatter plot of magnitude of the baroclinic feedback associated with the non-standard EOF structures in Fig. 7: x-axis: estimate from the LRF experiments using (10), y-axis: estimate from the IWP model. The line $y = x$ is green, the least squares line, $y = 1.05 \cdot x$ is blue

4. Baroclinic Feedback Estimates from Observations

In this section, we apply the IWP model to ERA-interim reanalysis (Dee et al. (2011)) data to estimate observed feedbacks. The analysis is performed for a 5-month "season" centered on each of the twelve calendar months. The mean EP fluxes from the IWP model deviate from observed EP fluxes in the deep tropics (not shown) due to tropical Rossby wave sources that are not captured by s and therefore (14) is not strictly valid. In the extratropics, however, the model captures the observed EP fluxes. We have also attempted to apply RWC to observations, however, the simulation of the climatological $\overline{u'v'}$ is poor, perhaps due to zonal asymmetries in the U climatology.

To estimate the eddy fluxes "caused" by U anomalies, we first use time-lagged regression analysis between the leading U PC and eddy wave activity fluxes (Lorenz and Hartmann (2001)). Next, the eddy fluxes are averaged from lags 10-15 days after the peak in the U PC. This defines the eddy "response" to the U EOF in our analysis. Following the RWC methodology, the EP flux budget is then averaged from 500-100hPa. In Fig. 10a, we show the eddy momentum flux convergence, m, response to EOF1 as a function of time of year (5-month "seasons" centered on each month). Two analyses, one for each hemisphere, are joined at the equator and superimposed on the same plot. The response is largest in Northern Hemisphere winter, consistent with Lorenz and Hartmann (2003), and smallest in Northern Hemisphere summer. In the Southern Hemisphere, the m response

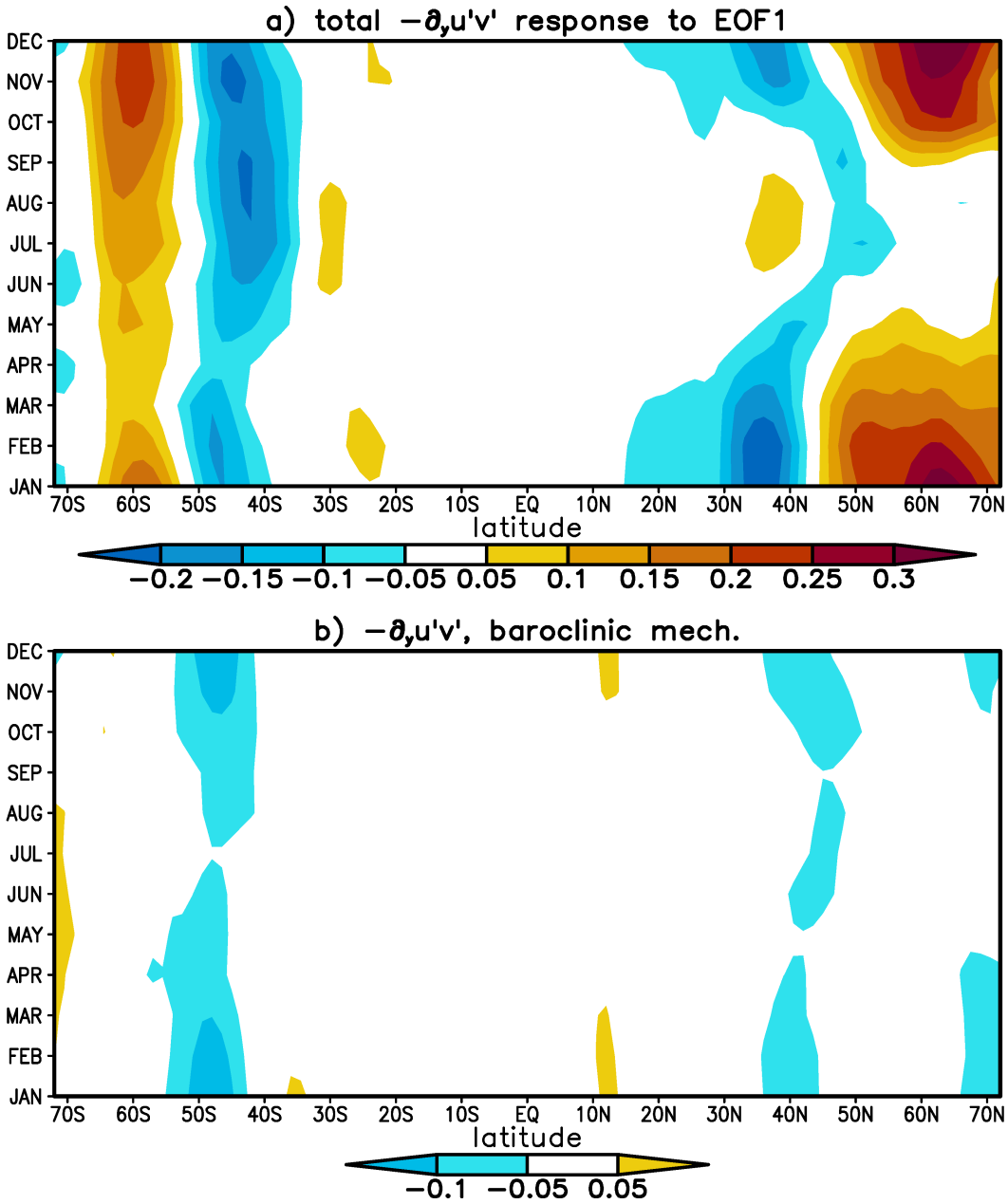

FIG. 10. a) Response of eddy momentum flux convergence to EOF1 estimated from lagged regressions (m/s/day). x-axis is latitude and y-axis is month. The panel shows results of two separate lagged regressions-one for the Southern Hemisphere and one for the Northern Hemisphere-that are joined together at the equator. The eddy momentum flux convergence is averaged over time lags 10-15 days after the peak in U. b) Same as (a) but for the portion of the eddy momentum flux convergence from the baroclinic feedback (estimated from the IWP model).

is largest in November and December, which is consistent with the seasonal cycle of annular mode persistence found in Baldwin et al. (2003).

The baroclinic feedback estimated from the IWP model is shown in Fig. 10b. Like the two-level GCM (Fig. 8c), the baroclinic feedback accounts for a portion of the equatorward center of action of m and practically none of the poleward center of action. In the Northern Hemisphere, the latitude of the baroclinic feedback varies much less through the seasonal cycle than the total m. Due to the strong zonal asymmetries in the Northern Hemisphere, it is possible that the zonal-mean picture ignores important details of the structure of the baroclinic feedback. This might also be true in Southern Hemisphere winter. In future work, we will explore generalizing the IWP in the longitude direction using the EP flux of Plumb (1986).

Projecting the baroclinic feedback on the total m and normalizing by the m amplitude gives its relative contribution to the total (Fig. 11). The barotropic feedback is projected on the total m in the same way. The residual (gray line) is simple one minus the baroclinic and barotropic contribution. First, one should focus on the solid lines only. During some time periods, for example Northern Hemisphere winter, the contribution of the baroclinic feedback is zero. Evidently, the baroclinic feedback structure is orthogonal to the total m. The baroclinic feedback makes the largest contribution to m in March in the Southern Hemisphere and June in the Northern hemisphere where it approaches or exceeds 40% of the total. In most cases, however, the baroclinic feedback in observations is weaker than that in the two-level primitive equation model. In many cases, the barotropic feedback accounts for the remaining feedback, however, in spring and summer of the Northern Hemisphere the residual is important. Using the anomalous wave activity budget in isolation, Nie et al. (2014) found that the baroclinic feedback is about 50% of the barotropic feedback in the Southern Hemisphere. Our modified estimates, which consider wave propagation efficiency, suggest the baroclinic feedback is even smaller with 9 of 12 months having a baroclinic feedback less than or equal to 25% of the barotropic feedback.

In the Southern Hemisphere, Nie et al. (2014) found that the baroclinic feedback becomes dominant if one considers the 100-250hPa average for the wave activity budget rather than 100-500hPa. This is also true in our IWP analysis for the Southern Hemisphere only (not shown). The variability of U, however, depends on the vertically integrated eddy momentum flux and 100-250hPa accounts for less than half of the total for 100-500hPa. Apparently, the barotropic feedback

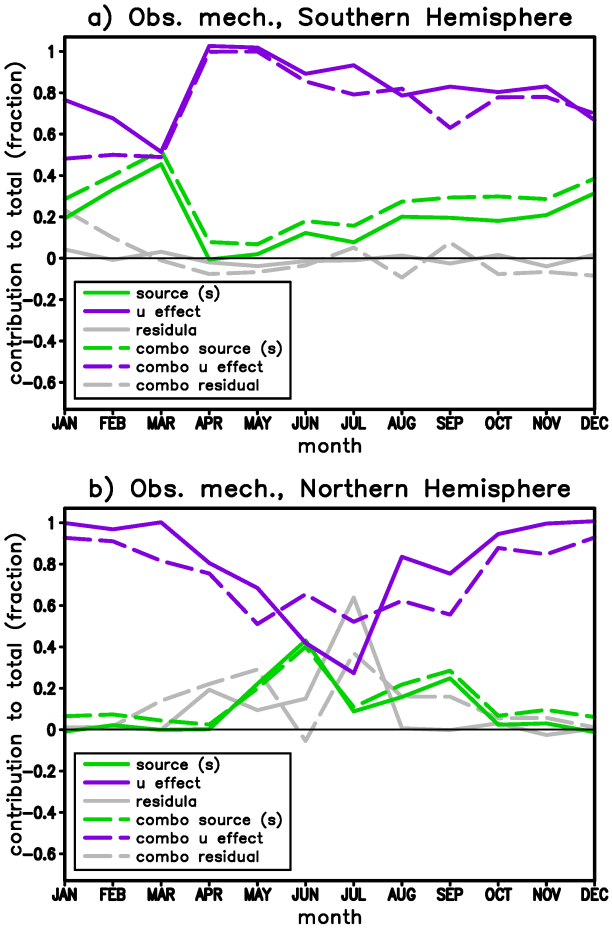

FIG. 11. Contribution of the baroclinic (green) and barotropic (purple) feedback to the total eddy momentum flux convergence response as a function of month. The residual (gray) is 1 minus the sum of the baroclinic and barotropic effect. The dashed lines represent the combined (summed) contribution of two separate IWP analyses: one for the 100-250hPa layer and one for the 250-500hPa layer (see text). a) Southern Hemisphere. b) Northern Hemisphere.

for 250-500hPa is large enough to offset the baroclinic feedback in 100-500hPa. This is confirmed by calculating a separate IWP model to the 100-250hPa and 250-500hPa layers and then adding the results (dashed lines in Fig 11). This "combination" analysis is like the single IWP applied 100-500hPa. However, in general, we do not believe it is a good idea to apply the IWP to thin levels of the upper troposphere because Birner et al. (2013) find localized regions of up-gradient PV fluxes in this region, which is not consistent with the IWP assumption that baroclinic instability

is the only source of wave activity. Therefore, we believe it is best to average over the full depth of the upper troposphere.

5. Conclusions

In this paper, we provide multiple independent estimates of the relative role of baroclinic and barotropic feedbacks on the eddy response to EOF1 U anomalies in both models and reanalyses. The goal is to assess the validity of various methodologies used in the literature that suggest baroclinic feedbacks are dominant. All our estimates suggest that barotropic feedbacks are the most important (70% of total) but that baroclinic feedbacks are non-negligible (30%). The estimates come from the following methods/experiments: Rossby Wave Chromatography (RWC, Lorenz (2015) and Section 2c), Linear Response Function (LRF) methodology (Hassanzadeh and Kuang (2016)) and a new method we call Inferred Wave Propagation (IWP) modelling.

Using linearized dynamics, RWC calculates the eddy momentum flux from 1) the baroclinic wave activity source and 2) the meridional wave propagation characteristics of the background flow. By changing these two components separately, the relative roles of the baroclinic (component 1) and barotropic (component 2) feedback on U anomalies can be assessed. The structure of the momentum flux response to each component can also be calculated.

Hassanzadeh and Kuang (2019) argued for baroclinic mechanisms by manipulating the vertical structure of EOF1 using the LRF method. They found that a barotropic version of EOF1 did not have a positive eddy feedback, presumably because the baroclinic source does not change for barotropic anomalies. We repeat these experiments, however, and find that the baroclinic source response is not zero but strongly negative, damping the eddy feedback. Here, we start with the hypothesis that both the baroclinic and barotropic mechanisms are relevant. With two mechanisms and two cases (standard EOF and barotropic EOF) one can linearly disentangle the role of each mechanism. Consistent with RWC, we find that the barotropic mechanism is more important. We repeat this analysis for multiple alternative changes in the EOF1 vertical structure: for example, baroclinic U anomalies with zero surface U. All five perturbed EOF1 structures imply similar magnitudes of the barotropic and baroclinic feedback.

The LRF experiments also show that barotropic U anomalies lead to opposing, negative baroclinic source anomalies. Currently, we do not understand the dynamical reasons for such a response

although one potential explanation involves Gliatto and Held (2020), who find that Rossby waves propagating meridionally in a vertically and meridionally varying background flow can give rise to non-zero vertical EP flux. The theories on the suppression of baroclinic instability by barotropic shear (James (1987), Nakamura (1993)) do not appear to explain the baroclinic source changes because the barotropic shear changes are meridionally out-of-phase with the baroclinic source changes.

The final estimate of the relative role of baroclinic and barotropic feedbacks uses the upper-level wave activity budget of the mean state to quantitatively infer the degree of meridional wave propagation in the mean state. This method, which we call Inferred Wave Propagation (IWP) modelling, predicts the momentum flux given the baroclinic wave activity source assuming wave propagation characteristics are the same as the mean state. When applied to an anomalous wave source, IWP gives the momentum flux from the baroclinic feedback. IWP more precisely quantifies the ideas on "radiation efficiency" introduced by Robert et al. (2017). The baroclinic/barotropic feedback estimates from IWP agree with RWC and LRF and moreover, the spatial structures of the two feedback components are very similar to the baroclinic RWC method. IWP can also be applied to feedbacks diagnosed from observations using lagged regression analysis (e.g. Lorenz and Hartmann (2001)). Most times of year, the barotropic feedback is greater than or equal to the value from the simple GCM (70%) and in some cases the barotropic feedback accounts for almost all the feedback. During March in the Southern Hemisphere and June in the Northern Hemisphere the barotropic feedback is smaller, but it is always greater than the baroclinic feedback.

All diagnostics and experiments in this paper suggest that barotropic feedbacks are dominant over baroclinic feedbacks. This is consistent with the success of barotropic models in capturing the structures and feedbacks associated with EOF1 U anomalies (Vallis et al. (2004); Jin et al. (2006a); Jin et al. (2006b); Barnes and Hartmann (2011); Lorenz (2014b)). We believe that much of the apparent model differences in baroclinic feedback strength are due to the diagnostic methodologies used to attribute baroclinic feedbacks rather than true differences in feedback strength. For example, for the multi-level GCM used in this paper, the baroclinic feedback appears to account for all the feedback via an analysis of the anomalous wave activity budget. Further experiments that manipulate the friction acting on the zonal-mean zonal wind anomalies, however, show that most of the feedback is in fact barotropic. Feedback diagnostics based on the anomalous

wave activity budget are problematic if they do not take into account the efficiency of the meridional propagation of wave activity away from the jet (Robert et al. (2017)).

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Data availability statement. Please contact the author for the model code and data.

APPENDIX

Baroclinic RWC Methodology

In this appendix, we describe the methodology for prescribing the vertical EP flux in the linearized two-level QG model. First, the solutions to (5) are separated into individual zonal wavenumbers and phase speeds. Next, each of the resulting linear systems are written in matrix form:

$$\psi = \mathbf{AF}, \tag{A1}$$

where $\mathbf{A}$ is a matrix, ψ and $\mathbf{F}$ are vectors of complex eddy streamfunction and forcing amplitudes and we use simple second order differencing for the numerics. Note that $\mathbf{F}$ and F do *not* denote EP flux in this appendix. The vectors ψ and $\mathbf{F}$ are written with all latitudes of level 1 first and then all latitudes of level 2. As discussed in Lorenz (2015), using a vector to represent F assumes that F is coherent at all latitudes (i.e., at all times, F at high latitudes always has the same phase and amplitude relationship to F at low latitudes). Therefore, we follow Lorenz (2015) by adding a second dimension to F, which represents independent "realizations" of F. The individual realizations are analogous to the partitioning of data into separate chunks when doing cross-spectral analysis. The same realization argument holds for the streamfunction, ψ. Therefore, (A1) becomes a matrix equation:

$$\mathbf{\Psi} = \mathbf{AF}. \tag{A2}$$

Multiplying each side of (A2) by the complex conjugate transpose of itself:

$$\mathbf{\Psi} = \mathbf{ACA}^H, \tag{A3}$$

where $\mathbf{\Psi}$ and $\mathbf{C}$ are the ψ and F covariance matrices, respectively, and $\mathbf{A}^H$ denotes the conjugate transpose. We assume that $\mathbf{C}$ is real, which means that, on average, there is no net horizontal- or

vertical-phase tilt to the nonlinear forcing. Let n be the number of latitudes, then the covariance between the upper- and lower-level streamfunction is $\Psi_{j,n+j}$, where j is an index over latitude. The vertical EP flux is proportional to the imaginary part of $\Psi_{j,n+j}$:

$$\text{source} = \frac{mk_D^2}{2a}\Im(\Psi_{j,n+j}), \tag{A4}$$

where m is the integer zonal wavenumber and the factor of two comes from the fact that the zonal mean of two waves with complex amplitudes x and y is $xy^*/2$. Equation (A4) gives n constraints on the solution to (A3). In addition, we constrain the ratio of upper- to lower-level streamfunction amplitude:

$$\sum_j \left(p_{\text{upper}}\cos\phi\right)^2 = A\sum_j \left(p_{\text{lower}}\cos\phi\right)^2, \tag{A5}$$

where A is the ratio from the GCM and j is an index over latitude. There are a total of $n+1$ contraints. The problem is to find either $\mathbf{\Psi}$ or $\mathbf{C}$ from these contraints. Because of the assumption that $\mathbf{C}$ is real, it is most straightforward to treat $\mathbf{C}$ as the unknown variable subject to the $n+1$ contraints. Because $\mathbf{C}$ is symmetric, there are $\frac{n(n+1)}{2}$ unknowns. To close the problem, we choose the smallest forcing (= sum of diagonal elements of $\mathbf{C}$) that satisfies the constraints and is a valid covariance matrix (i.e. $\mathbf{C}$ is positive semidefinite). This is a "semidefinite programming" problem (Helmberg et al. (1996); Vandenberghe and Boyd (1996)) and multiple high quality libraries exist in multiple popular programming languages. Once the solution ($\mathbf{C}$) is found, the streamfunction is given by the diagonal elements of $\mathbf{\Psi}$ in (A3). To find the eddy momentum flux, one first creates a matrix that converts streamfunction to v and a matrix that converts streamfunction to u. One then right multiplies $\mathbf{\Psi}$ by the v matrix and left multiplies $\mathbf{\Psi}$ by the complex conjugate transpose of the u matrix. The eddy momentum flux is then $\frac{1}{2}$ times the real part of the diagonal elements of the resulting matrix.

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